Coding Theory :-

Consider the set $Z_2 = \{0, 1\}$ and the additive group $(Z_2, +)$ where $+$ denotes addition modulo 2. For any positive integer n we have.

$$Z_2^n = Z_2 \times Z_2 \times \cdots \times Z_2 \quad (n \text{ factors})$$

$$= \{(a_1, a_2 \ldots a_n) \mid (a_1, a_2 \ldots a_n \in Z_2\}$$

Hence every element of Z_2^n is an n-tuple $(a_1, a_2 \ldots a_n)$ in which each entry a_i belongs to Z_2 ie it is 0 or 1. Here n tuple is called a word or a string in Z_2^n and each a_i ($= 0$ or 1) in this word is called a bit (or component or an individual signal)

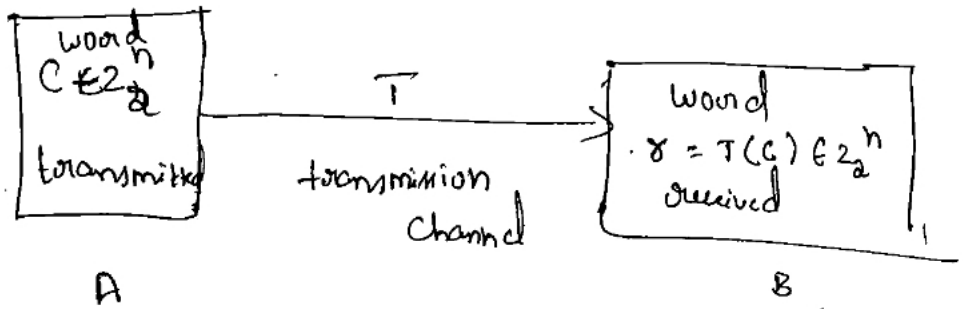
Ex:- 110 is a word that represents the triple $(1, 1, 0) \in Z_2^3$ and 00101 is a word that represents the 5-tuple $(0, 0, 1, 0, 1) \in Z_2^5$

NOTE:- The component wise addition of n-tuples Z_2^n is a group and it is denoted as $(Z_2^n, +)$ Here $+$ is the component wise addition

Transmission of a word :- Suppose a word $c = c_1 c_2 c_3 \ldots c_n \in Z_2^n$ is transmitted from a point A through a transmission channel T.

This word would be received at a point B without any change. But actually in Practice, transmission Channel suffer disturbances (generally noise) that may cause a 0 to be received as 1 and vice-versa.

Suppose the word c is transmitted from a point A is received as different word $\gamma \in Z_2^n$ at B the word γ will be of the form $\gamma = \gamma_1 \gamma_2 \ldots \gamma_n$ where each γ_i is 0 or 1 but γ_j to c_j for some j, $1 \le j \le n$.



If $\gamma_i = c_i$ for all values of i except ($k \le n$) then γ differs from c in k places. The word γ is denoted by $T(c)$.

i.e. $\gamma = c + e$ where e is word in Z_2^n

The word e is called Error Pattern.

Binary Symmetric Channel :-

Suppose P_1 is the Probability that the transmitted signal 0 is received as the signal 1 and P_2 is the Probability that the transmitted signal 1 is received as the signal 0. The transmitted channel is symmetric if

$$P_1 = P_2 = P_c$$

If P is the Probability, then the Probability that a signal (0 or 1) is received Correctly is $1-P$. Hence the individual signals being Considered for transmission and received (0 and 1) are in number, a symmetric Channel through which a signal 0 or 1 is transmitted is referred to as binary symmetric channel.

Probability :-

Unit-3

Coding Theory:-

Suppose γ differs from c in exactly one place (i.e j^{th} place). Then $\gamma_1 = c_1, \gamma_2 = c_2, \ldots,$
$\gamma_{j-1} = c_{j-1}, \gamma_j \neq c_j, \gamma_{j+1} = c_{j+1},$
$\gamma_{j+2} = c_{j+2}, \ldots, \gamma_n = c_n.$

Since the Probability that $\gamma_i = c_i$ is $(1-p)$ for each i, the Probability $\gamma_1 = c_1, \gamma_2 = c_2, \ldots$
$\gamma_{j-1} = c_{j-1}$ is

By Product rule
$$\underbrace{(1-p)(1-p)\ldots(1-p)}_{(j-1) \text{ factors}} = (1-p)^{j-1}$$

Next, the Probability that $\gamma_{j+1} = c_{j+1}, \ldots$
$\gamma_{j+2} = c_{j+2} \ldots, \gamma_n = c_n$ is

$$\underbrace{(1-p)(1-p)\ldots(1-p)}_{(n-j) \text{ factors}} = (1-p)^{n-j}$$

The Probability that $\gamma_j \neq c_j$ is p.

$\therefore$ The Probability that $\gamma_1 = c_1, \gamma_2 = c_2, \ldots$
$\gamma_{j-1} = c_{j-1}, \gamma_j \neq c_j, \gamma_{j+1} = c_{j+1}$
$\ldots \gamma_n = c_n$ by Product is,
$(1-p)^{j-1} \cdot p \cdot (1-p)^{n-j} = p(1-p)^{n-1}$

Hence this is the Probability that r differs from c in exactly one place.

Similarly, the Probability that r differs from c in k places is

$$p^k (1-p)^{n-k}$$

Suppose that k Errors are made in the transmission of C if the received word r differs from C in k places. Since C has n places and k of these n places can be choosen in ${}^nC_k$ number of ways. the Probability that k Errors are made in the transmission of C is

$$\boxed{{}^nC_k \cdot p^k (1-p)^{n-k}}$$

(1) The word $C = 1010110$ is transmitted through a binary Symmetric channel. If $e = 0101101$ is the Error Pattern, find the word or received r?

$p = 0.05$ is the Probability that a signal is incorrectly received, find the Probability with which r is received.

→ Transmitted word $C = 1010110 \in Z_2^7$

Error Pattern $C = 0101101 \in Z_2^7$

$\therefore$ Recived word $r = C + e$ where $+$ is the addition in Z_2^7.

$C = 1010110$

$= (1, 0, 1, 0, 1, 1, 0)$

$C = 0101101$

$= (0, 1, 0, 1, 1, 0, 1)$

$\gamma = C + C = (1, 0, 1, 0, 1, 1, 0) + (0, 1, 0, 1, 1, 0, 1)$

$= (1+0, 0+1, 1+0, 0+1, 1+1, 1+0, 0+1)$

$= (1, 1, 1, 1, 0, 1, 1) \in 2_2$

γ differs from C in the second, fourth, fifth and seventh places in totally 4 places.

∴ Probability with which γ received is

$P^4 (1-P)^{7-4} = (0.05)^4 (1-0.05)^3$

$= (0.05)^4 (0.95)^3$

$= 0.000005$

~~Generator Matrix~~ : Parity ~~Check Matrix~~.

(2) The word $C = 1010110$ is sent through a binary symmetric channel. If $p = 0.02$ is the Probability of incorrect receipt of a signal, find the Probability that c is received as $r = 1011111$. Determine the Error Pattern.

Soln:- Here, The transmitted word is $c = 1010110 \in Z_2^7$

The received word is $r = 1011111 \in Z_2^7$.

By Comparing c and r, we find that these differ in two places (that is fourth and seventh)

∴ The Probability that c is received as r is

$$p^k (1-p)^{n-k}$$

$$p^2 (1-p)^{7-2}$$

$$= (0.02)^2 (1-0.02)^5$$

$$= (0.02)^2 (0.98)^5$$

$$= 0.00036.$$

The Error Pattern e is given by $r = c + e$.

If we have set $e = e_1 e_2 e_3 \cdots e_7$

we should have $r = c + e_1 e_2 e_3 \cdots e_7$ that is

$$1011111 = 1010110 + e_1 e_2 e_3 \cdots e_7$$

Thus, in view of the Component-wise addition rule, we have

$$1 = 1 + e_1 \text{ so that } e_1 = 0$$
$$0 = 0 + e_2 \text{ --- } e_2 = 0$$
$$1 = 1 + e_3 \text{ --- } e_3 = 0$$
$$1 = 0 + e_4 \text{ --- } e_4 = 1$$
$$1 = 1 + e_5 \text{ --- } e_5 = 0$$
$$1 = 1 + e_6 \text{ --- } e_6 = 0$$
$$1 = 0 + e_7 \text{ --- } e_7 = 1$$

$$c = e_1 e_2 e_3 e_4 e_5 e_6 e_7 = \underline{0001001}$$

(3) A binary symmetric channel has Probability $P = 0.05$ of incorrect transmission. If the word $C = 01101110$, 1 is transmitted, what is the Probability that (i) single Error occurs, (ii) a double Error occurs (iii) a triple Error occurs, and (iv) three Errors occur no two of them Consecutive?

Soln:- Given $C = 011011101 \in Z_2^9$

(i) The Probability that there occurs a single Error in transmission is

$$({}^9C_1) P (1-P)^{9-1}$$

$$= 9 \times 0.05 \times (0.95)^8$$

$$= \underline{0.2985}$$

(ii) The Probability that a double error occurs is

$$(9C_2)\,p^2(1-p)^{9-2}$$

$$= 36 \times (0.05)^2 \times (0.95)^7$$

$$= 0.06285$$

(iii) The Probability that a triple error occurs is

$$(9C_3)\,p^3(1-p)^{9-3}$$

$$= 84 \times (0.05)^3 \times (0.95)^6$$

$$= 0.0077185$$

(iv) The Probability that three Errors occurs no two of which are Consecutive is $P_1 + P_2$, where P_1 is the Probability that Error occurs in any three of the first, third, fifth, seventh and ninth Places and P_2 is the Probability that Error Occurs in any three of the second, fourth sixth and eight-places.

$$P_1 = (5C_3)\,p^3(1-p)^{5-3}$$

$$= 10 \times (0.05)^3 \times (0.95)^2$$

$$= 0.00112812\overline{5}$$

and

$$P_2 = (4C_3)\,p^3(1-p)^{4-3}$$

$$= 4 \times (0.05)^3 \times (0.95)$$

$$= 0.0016031\overline{25}$$

$\therefore$ ~~$P_1 + P_2 = 0.00112812\overline{5} + 0.0016031\overline{25}$~~

∴ The required Probability is

$P_1 + P_2 = 0.001128125 + 0.000475$

$= 0.00160325$

① An encoding function $E : Z_2^8 \to Z_2^9$ is defined as follows. For each $w = w_1, w_2 \ldots w_8 \in Z_2^8$

$E(w) = w_1 w_2 \ldots w_8 w_9$, where $w_9 = \sum_{i=1}^{8} w_i$, the summation being taken under modulo 2 addition.

If $p = 0.001$ is the Probability that a signal is received incorrectly, find the Probability that the code word 110101101 is received with at most one Error.

Soln:- Consider the word $w = w_1 w_2 \ldots w_8$. By using the given definition of w_9, we note that if an odd number of w_is in w are 1's, then $w_9 = 1$

In this case $C = E(w) = w_1 w_2 \ldots w_8 w_9$ contains even number of 1's.

or

if an even no of w_is in w are 1's, then $w_9 = 0$

Hence $C = E(w)$ contains an even number of 1's

The given ~~code~~ encoding function the code word $C = E(w)$ of every word $w \in Z_2^8$ contains an even number of 1's.

Consider code word $c = 110101101 \in Z_2^9$.

If p is the Probability that a signal is incorrectly received.

Then the Probability of receiving c correctly is $(1-p)^9$.

Probability of receiving c with one Error is

$$(9C_1).p(1-p)^8$$

∴ The Probability of receiving c with at most one Error is

$$(1-p)^9 + (9C_1).p(1-p)^8$$

$$(1-0.001)^9 + 9 \times 0.001 \times (1-0.001)^8$$

$$= (0.999)^9 + 0.009 \times (0.999)^8$$

$$= 0.999964167$$

$\therefore c = 0001001$

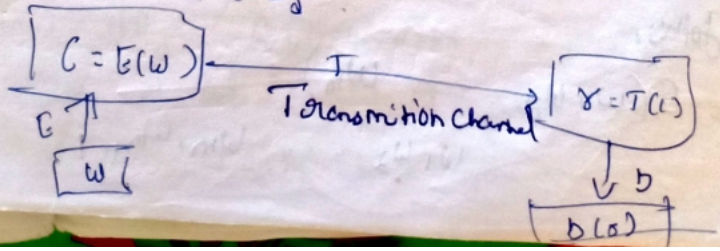
Encoding and Decoding message:

Suppose a set of words has to be transmitted each containing m components (bits). Let W denote the set of components i.e. $W \subseteq Z_2^m$ While transmitting the word the number of error has to be reduced.

Let $n > m$ and to each $w \in W$, assign a unique word $C \in Z_2^n$. The assignment is carried out by appending $n-m$ extra components (0's and 1's) to w through a function (also to one) $E: W \to Z_2^n$ so that $C = E(w)$.

The extra 0's and 1's appended to w is to detect or correct error produced in the transmission channel. The process of appending extra 0's and 1's to w through the function E is called the Encoding process

The word $C = E(w) \in Z_2^n$ is transmitted by means of transmission channel. Let $T(C) = \gamma \in Z_2^n$ be the word received In γ, the extra components contained there in are removed through an onto function $D: Z_2^n \to Z_2^m$ and obtain a word $D(\gamma) \in Z_2^m$



If the transmission channel is noiseless, then the word $D(\gamma)$ received at the end of the process is identified identical with the original word w. This is not possible in a practical situation. Hence the function E and D are so chosen that there is a high probability for $D(\gamma)$ to be identical with w (The process of recapturing the original word w from $D(\gamma)$ with this high probability is called the decoding process.

The word $w \in W \subseteq Z_2^m$ that is chosen for transmission is called message. The word $c = E(w) \in Z_2^n$ is called the Code word assigned to w. The set $C = E(W) = \{E(w) / w \in W\} \subseteq Z_2^n$ is called the set of codes or just the code. The word $\gamma = T(c)$ is called the received word and $D(\gamma)$ is called decoded word. The ratio m/n measures is called the efficiency of the coding scheme and is called the ~~ratio of code~~ rate of the code. The Function E is called Encoding function of an (n, m) block code. The function D is called the decoding function of an (n, m) block code.

Problem :- Suppose the encoding function $E: Z_2^m \to Z_2^{m+1}$ is defined as follows

$$E(w) = E(w_1 w_2 \ldots w_m) = w_1 w_2 \ldots w_m w_{m+1}$$

(2) Suppose the encoding function $E: Z_2^m \to Z_2^{m+1}$ is defined as follows.

$$E(w) = E(w_1, w_2, \ldots w_m) = w_1 w_2 \ldots w_m w_{m+1},$$

where $w_{m+1} = \begin{cases} 0 & \text{if } w \text{ contains even no 1's} \\ 1 & \text{if } w \text{ contains odd no 1's} \end{cases}$

and the corresponding decoding function $D: Z_2^{m+1} \to Z_2^m$ is $D(a) = D(a_1 a_2 \ldots a_m a_{m+1}) = a_1 a_2 \ldots a_m$.

(a) Find the code words assigned by E to the following messages in Z_2^3.

000, 001, 011, 100, 110, 101, 111, 010

b) Find the decoded words assigned by D to the following received words in Z_2^4.

0000, 0001, 0101, 1111, 1010, 1100, 1101, 1001.

Soln:-

(a) By using the definition of E, we find that.

$E(000) = 0000$ $E(001) = 0011$
$E(011) = 0110$ $E(100) = 1001$
$E(110) = 1100$ $E(101) = 1010$
$E(111) = 1111$ $E(010) = 0101$

(b)

(b) By using the definition of D, we find that.

$D(0000) = 000$ $\quad D(0001) = 000$
$D(0101) = 010$ $\quad D(1111) = 111$
$D(1010) = 101$ $\quad D(1100) = 110$
$D(1101) = 110$ $\quad D(1001) = 100$

(3) Suppose the encoding function $E: Z_2^m \to Z_2^{3m}$ and the corresponding decoding function $D: Z_2^{3m} \to Z_2^m$ are as follows.

$E(w) \equiv E(w_1 w_2 \ldots w_m) = w_1 w_2 \ldots w_m, w_1 w_2 \ldots w_m, w_1 w_2 w_3 \ldots w_m$

$D(a) \equiv D(a_1 a_2 \ldots a_m a_{m+1} \ldots a_{2m} a_{2m+1} \ldots a_{3m}) = s_1 s_2 \ldots s_m$

where s_i is given by the majority rule, namely

$$s_i = \begin{cases} 1 & \text{if } a_i a_{i+m} a_{i+2m} \text{ has a majority of 1's} \\ 0 & \text{if } a_i a_{i+m} a_{i+2m} \text{ has a majority of 0's} \end{cases}$$

(a) Find the code words assigned by E to the following messages in Z_2^3

000, 01, 010, 100, 001, 101, 110, 111

(b) Find the decoded words assigned by D to the following received words in Z_2^6.

111111, 101010, 010101, 100100
010011, 110110, 010111, 000111

Soln:- (a) For $w = w_1w_2w_3 \in Z_2^3$ - the given definition of E yields ($E(w) = w_1w_2w_3\, w_1w_2w_3\, w_1w_2w_3$

Thus we have.

$E(000) = 000000000$
$E(011) = 011011011$
$E(010) = 010010010$
$E(100) = 100100100$
$E(001) = 001001001$
$E(101) = 101101101$
$E(110) = 110110110$
$E(111) = 111111111$

(b) For $x = x_1x_2x_3x_4x_5x_6 \in Z_2^6$, the given definition of D yields (on $n = 2$)

$$D(x) = s_1s_2$$

where

$$s_1 = \begin{cases} 1 & \text{if } x_1x_3x_5 \text{ has a majority of '1'} \\ 0 & \text{if } x_1x_3x_5 \text{ has a majority of '0'} \end{cases}$$

and

$$s_2 = \begin{cases} 1 & \text{if } x_2x_4x_6 \text{ has a majority of '1'} \\ 0 & \text{if } x_2x_4x_6 \text{ has a majority of '0'} \end{cases}$$

we find that

$D(111111) = 11$ $\quad D(101010) = 10$
$D(010101) = 01$ $\quad D(100100) = 00$
$D(010011) = 01$ $\quad D(110110) = 11$
$D(010111) = 01$ $\quad D(000111) = 01$

The Hamming matrix:-

Consider a word $x \in Z_2^m$. Then $x = x_1 x_2 \cdots x_m$ where each x_i is either a 0 or a 1. The number of x_i's in x which are 1's is called the weight of x and is denoted by $|x|$ or $wt(x)$. The weight $wt(x)$ is a non-negative integer.

For ex:-

if $x = 1010011 \in Z_2^7$

$|x| = wt(x) = 4$

if $y = 111100011101 \in Z_2^{12}$

$|y| = wt(y) = 8$.

Let $x = x_1 x_2 \cdots x_m$ and $y = y_1 y_2 \cdots y_m$ be words in Z_2^m. Then the number of i's such that $x_i \neq y_i$, $1 \leq i \leq m$ is called the Hamming distance between x and y and is denoted by $d(x,y)$. In other words, $d(x,y)$ is the number of positions in which x and y differ.

Ex:- if $x = 01001$, $y = 11010$

x and y differ in 3 places.

$\therefore d(x,y) = 3$

$x + y = 10011$

$wt(x+y) = 3$

Hamming metric :-

When a function $d = d(x,y)$ of two variables x and y satisfies the properties

$d(x,y) = d(y,x)$

$d(x,y) \geq 0$

$d(x,y) = 0$ iff $x = y$

$d(x,z) \leq d(x,y) + d(y,z)$ Then

$d(x,y)$ is a metric or distance function.

$\therefore d = d(x,y)$ is known as Hamming metric

minimum distance :-

Let $E : W \to Z_2^n$, where $W \subseteq Z_2^m$, $m < n$ be an encoding function, we compute the distance $d(E(x), E(y))$ for all $x, y \in W$. The minimum of these distance is called the minimum distance between the code words in $C = E(W)$ (or simply, the minimum distance in E) and is denoted by $\min(E)$.

Thus

$$\min(E) = \min\{d(E(x), E(y)) \mid x, y \in W\}$$

For Ex:- $W = Z_2^2 = \{00, 10, 01, 11\}$

and $E : Z_2^2 \to Z_2^6$ be the encoding function of the (6,2) triple repetition code Then.

$E(00) = 000000$ $E(10) = 101010$

$E(10) = 101010$ $E(11) = 111111$

Hamming metric :-

When a function $d = d(x,y)$ of two variables x and y satisfies the properties

$d(x,y) = d(y,x)$

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For Ex :- $w = z_2^2 = \{00, 10, 01, 11\}$

and $E : z_2^2 \to z_2^6$ be the encoding function of the $(6,2)$ triple repetition code Then.

$E(00) = 000000$ $E(10) = 101010$

$E(10) = 101010$ $E(11) = 111111$

Soln:-

$$d(\bar{E}(00), \bar{E}(10)) = 3$$

$$d(\bar{E}(00), \bar{E}(01)) = 3$$

$$d(\bar{E}(00), \bar{E}(11)) = 6$$

$$d(\bar{E}(10), \bar{E}(01)) = 6$$

$$d(\bar{E}(10), \bar{E}(11)) = 3$$

$$d(\bar{E}(01), \bar{E}(11)) = 3$$

$\therefore \min(\bar{E}) = 3$

$= 0.000005$

Generator Matrix : Parity Check Matrix.

Error detection and Error Correction.

Theorem (Error detection)

Let $E: w \to c$ be an Encoding function where $w \subseteq 2_2^m$ and $C = E(w) \subseteq 2_2^n$ $m < n$ for any $k \in 2^+$, we can detect Errors of weight less than or equal to k iff the minimum distance in E is atleast $k+1$

Theorem : (Error Correction)

Let $E : W \to C$ be an Encoding function, where $W \subseteq Z_2^m$ and $C = E(W) \subseteq Z_2^n$, $m < n$. Then There Exists a decoding function that Corrects all Transmission Errors of weight less than or Equal to k iff $\min(E)$ is atleast $2k+1$.

Problem:-

Let $W = Z_2^2$ and $E : Z_2^2 \to Z_2^8$ be an Encoding function with minimum distance 5. Using this Code, how many Errors can we detect and then how many Errors can we Correct?

$\longrightarrow$ The given Code can detect Errors of weight $\leq S-1$ where $S = \min(E)$. Since $S = 5$ we can detect all Errors of weight ≤ 4 (By Theorem of Error detection)

We can Correct all Errors of weight $\leq \frac{1}{2}(S-1)$ where $S = \min(E) = 5$.

$\therefore$ Error can be Corrected of weight ≤ 2.

(Theorem of Error Correction)

(2) For each of the following Encoding functions, find the minimum distance blw the code words. Indicate the Error detecting and Error-Correcting Capabilities of each code.

(i) $E: Z_2^2 \longrightarrow Z_2^5$ defined by.

$E(00) = 00001$ $\quad E(01) = 01010$

$E(10) = 10100$ $\quad E(11) = 11111$

Soln:- $d(E(00), E(01)) = 3$ $\quad d(E(00), E(10)) = 3$

$d(E(00), E(11)) = 4$ $\quad d(E(01), E(10)) = 4$

$d(E(01), E(11)) = 3$ $\quad d(E(10), E(11)) = 3$

$\min(E) = 3$

The Encoding function detects all errors of weight $\leq (s-1)$ where $s = 3$

i.e $(3-1) = 2$.

The code will correct Errors of weight $\leq \frac{1}{2}(s-1) = \leq \frac{1}{2}(3-1) = 1$

i.e the code can correct all single Errors.

ii) $E: Z_2^3 \longrightarrow Z_2^6$ defined by

$E(000) = 000111$ $E(001) = 001001$

$E(010) = 010010$ $E(011) = 011100$

$E(100) = 100100$ $E(101) = 101010$

$E(110) = 110001$ $E(111) = 111000$

n k

Soln:- $d(E(000), E(001)) = 3$ $d(E(000), E(010)) = 3$

$d(E(000), E(011)) = 4$ $d(E(000), E(100)) = 3$

$d(E(000), E(101)) = 4$ $d(E(000), E(100)) = 3$

$d(E(000), E(111)) = 6$ $d(E(000), E(110)) = 4$

$d(E(001), E(011)) = 3$ $d(E(001), E(010)) = 4$

$d(E(001), E(101)) = 3$ $d(E(001), E(100)) = 4$

$d(E(001), E(111)) = 3$ $d(E(001), E(110)) = 3$

$d(E(010), E(100)) = 4$ $d(E(010), E(011)) = 3$

$d(E(010), E(110)) = 4$ $d(E(010), E(111)) = 3$

$d(E(011), E(100)) = 3$ $d(E(011), E(101)) = 4$

$d(E(011), E(110)) = 4$ $d(E(011), E(111)) = 2$

$d(E(100), E(101)) = 3$ $d(E(100), E(110)) = 3$

$d(E(100), E(111)) = 3$ $d(E(101), E(110)) = 4$

$d(E(101), E(111)) = 2$ $d(E(110), E(111)) = 2$

$\min(E) = 2$

Code can detect errors of weight $\leq (2-1)$

$= 1$

and can correct all errors of weight $\leq \frac{1}{2}(2-1)$

$\leq \frac{1}{2}$

⇒ Code Corrects single errors and no Correction Capability.

Generator Matrix: Parity check matrix

Let I_m be the unit matrix of order m and A be the matrix of order $m \times (n-m)$ where m and n are positive integers defined over Z_2.

Let $G = [I_m \mid A]$ be the corresponding augmented matrix.

If $A = [a_{ij}]$, then G is the $m \times n$ matrix given by

$$G = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & a_{11} & a_{12} & \cdots & a_{1k} \\ 0 & 1 & 0 & \cdots & 0 & a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & & & & & & & & \\ 0 & 0 & 0 & \cdots & 1 & a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} \rightarrow (1)$$

where $k = n - m$.

Consider a word $w = w_1, w_2, w_3 \ldots w_m \in Z_2^m$

Let $[w]$ denote the row matrix such that

$$[w] = [w_1, w_2, w_3 \cdots w_m] \rightarrow (2)$$

Matrix Product $[w]G$ is

$$[w]G = [w_1, w_2, w_3 \ldots w_m] \begin{bmatrix} 100\ldots0 & a_{11} a_{12} \ldots a_{1k} \\ 010\ldots0 & a_{21} a_{22} \ldots a_{2k} \\ \vdots & \vdots \\ 000\ldots1 & a_{m1} a_{m2} \ldots a_{mk} \end{bmatrix}$$

$$[w]G = [w_1, w_2, w_3 \ldots w_m\ w_{m+1}, w_{m+2} \ldots w_n] \longrightarrow (3)$$

Where

$$w_{m+1} = w_1 a_{11} + w_2 a_{21} + w_3 a_{31} + \ldots + w_m a_{m1}$$

$$w_{m+2} = w_1 a_{12} + w_2 a_{22} + w_3 a_{23} + \ldots + w_m a_{m2}$$

$$\vdots$$

$$w_{m+k} = w_n = w_1 a_{1k} + w_2 a_{2k} + w_3 a_{3k} + \ldots + w_m a_{mk}$$

or

$$w_{m+i} = w_1 a_{1i} + w_2 a_{2i} + w_3 a_{3i} + \ldots + w_m a_{mi}$$

$$= \sum_{j=1}^{m} w_j a_{ji},$$

Where $i = 1, 2, 3 \ldots k = n-m.$ $\longrightarrow (4)$

Let C denotes the word whose components are the Elements of the row matrix $[w]G$.

i.e $C = w_1, w_2, w_3 \ldots w_m w_{m+1} w_{m+2} \ldots w_n$ $\longrightarrow (5)$

$C \in Z_2^n$. Consider the encoding function

$E: Z_2^m \longrightarrow Z_2^n$ defined as

$E(w) = C$ $\longrightarrow (6)$

ie $[E(w)] = [C]$ - ie matrix form → (7)

where $[E(w)]$ is the matrix whose elements are the components of $E(w)$ and $[C]$ is the matrix whose elements are the components of C.

From eq (5) & (3)

$$[C] = [w_1, w_2, w_3 \ldots w_n] = [w]G.$$

From eq (7)

$$[E(w)] = [w]G \rightarrow (8)$$

∴ For any $w \in Z_2^m$, $E(w)$ is uniquely determined by the matrix G,

∴ matrix G is called a Generator matrix of E.

Encoding scheme described above, the code word $C = E(w) = w_1, w_2 \ldots w_m w_{m+1} \ldots w_n$ assigned to a word $w = w_1, w_2 \ldots w_m$ is got by appending $(n-m)$ components $w_{n+1} \ldots w_n$ to w these $(n-m)$ components are determined through the system of eq (4). These eqn (4) are called the Parity - Check Equations.

Since $w_i = -w_i$ in Z_2

Eqn (4) can be written as

$$\sum_{j=1}^{m} w_j a_{ji} + w_{m+i} = 0 \longrightarrow (4')$$

for $i = 1, 2, \ldots k = n-m$

Consider the matrix H defined by

$$H = [A^T \mid I_{n-m}] \longrightarrow (9)$$

where A^T is the transpose of the matrix A and I_{n-m} is the unit matrix of order $n-m = k$. Since A is an $m \times (n-m)$ matrix, A^T is a $(n-m) \times m$ matrix. This matrix is determined by matrix A or matrix G and it is called the Parity check matrix associated with G.

Decoding Scheme :-

The Parity check matrix H provides a decoding scheme that corrects single errors in transmission if the following conditions hold.

(i) H does not contain a column of 0's

(ii) No two columns of H are identical.

When H satisfies these two conditions, the following algorithm is used to decode the received word

$r = r_1 r_2 \ldots r_n \in Z_2^n$.

(1) Find $H[r]^T$, where $[r]$ is the row matrix whose elements are $r_1, r_2, r_3 \ldots r_n$. If the matrix $H[r]^T$, called the syndrome of r, is a zero matrix, then the decoded message consists of the first m components of r.

(2) If $H[r]^T$ is equal to the i^{th} column of H, change the i^{th} component of r (from 1 to 0 or vice versa) in order to get the code word c. The first m components of c yield the original message.

(3) If neither of the cases (1) & (2) occurs, decoding cannot be done in reliable way.

(1) : An encoding function $E: Z_2^2 \longrightarrow Z_2^5$ is given by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

(a) Determine all the code words. What can be said about the error-detection capability of this code? What about its Error-Correction capability?

(b) Find the associated Parity Check matrix H.

(c) Use H To decode the received words 11101, 11011.

Soln:

G_r is of the form $G_r = [I_2 \mid A]$ where

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

(a) we find that.

$$[E(00)]\,G_r = [00]\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [00000]$$

$$[E(01)]\,G_r = [01]\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [01011]$$

$$[E(10)]\,G_r = [10]\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [10110]$$

$$[E(11)]\,G_r = [11]\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [11101]$$

These matrix eqn gives the code words.

$E(00) = 00000$, $E(01) = 01011$, $E(10) = 10110$

$E(11) = 11101$.

From this, we find that.

$d(E(00), E(01)) = 3$ $d(E(01), E(11)) = 3$

$d(E(00), E(10)) = 3$ $d(E(10), E(11)) = 3$.

$d(E(00), E(11)) = 4$

$d(E(01), E(10)) = 4$

$\therefore \min(E) = 3$.

$\therefore$ Code can detect all errors of weight $\leq (\delta - 1) \leq 2$ and can correct all single errors $\frac{1}{2}(\delta - 1) = \frac{1}{2}(3-1) = 1$.

(b) The Parity Check matrix H associated with G is given by:

$$H = [A^T | I_3] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

We observe that H does not contain a column of 0's & further no two columns of H are identical.

$\therefore$ H Corrects single Errors in transmission.

(c) For $r = 11101$, The syndrome of r is

$$H[11101]^T = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since this is a zero matrix, the decoded message is got by retaining the first two components of r. The decoded message is therefore 11.

For $r = 11011$, the syndrome of r is

$$H[11011]^T = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

we observe that this matrix is identical with the first column of H. ∴ we can change the first component of or from (1100) to get 01011. This is the code word.

The first two components of this code word, ie 01, is the original message.

(2) The generator matrix for an encoding function $E: Z_2^3 \to Z_2^6$ is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

(a) Find the code words assigned to 110 and 010

(b) obtain the associated Parity-Check matrix.

(c) Hence decode the received words 110110, 111101.

(d) Show that the decoding of 111111 is not Possible by using H.

Soln:- G is of the form.

$G = [I_3 | A]$, where

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

(a) $E(110) = [110] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} =$

$= [110101]$

and $[E(010)] = [010]\begin{bmatrix} 1&0&0&1&1&0 \\ 0&1&0&0&1&1 \\ 0&0&1&1&0&1 \end{bmatrix}$

$= [010011]$

Thus, the required Code words are:

$E(110) = 110101$ and $E(010) = 010011$

(b) The Parity-Check matrix associated with G is

$$H = [A^T | I_3] = \begin{bmatrix} 1&0&1&1&0&0 \\ 1&1&0&0&1&0 \\ 0&1&1&0&0&1 \end{bmatrix}$$

(c) For $r = 110110$, the Syndrome of r is

$$H[110110]^T = \begin{bmatrix} 1&0&1&1&0&0 \\ 1&1&0&0&1&0 \\ 0&1&1&0&0&1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

We observe that this matrix is identical with the second column of H. ∴ we change the second component in r (from 1100) to get the Code word $C = 100110$. The first three components of this Code word gives the original message $w = 100$.

For $r = 111101$, the syndrome of r is.

$$H[111101]^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

We observe that this matrix is identical with the third column of H. $\therefore$ we change the third component in r (from 1 to 0) to get the code word $c = 110101$. The first three components of this code word gives the original message $w = 110$

(d) For $r = 111111$, the syndrome of r is.

$$H[111111]^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

We observe this matrix is not a zero matrix and is not Equal to any column of H.

$\therefore$ we cannot decode $r = 111111$ by using H.

(3) The Parity-Check matrix for an Encoding function $E: Z_2^3 \longrightarrow Z_2^6$ is given by

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(a) Determine the associated generator matrix.

(b) Does this code Correct all single Errors in transmission?

(a) We have

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

which is of the form $[A^T | I_3]$. Accordingly,

$$A^T \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ so that } A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Hence, the associated generator matrix is.

$$\therefore G = [I_3 | A] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

(b) We observe that two columns of H (2^{nd} & 5^{th}) are identical.

$\therefore$ H does not Provide a decoding scheme that corrects Single Errors in Transmission.

(2) Define the Encoding function $E: Z_2^3 \longrightarrow Z_2^6$ by means of the Parity Check matrix.

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(a) Determine all code words.

(b) Does this Code Correct all Single Errors in transmission.

$$H = [A^T | I_3]$$

Where.

$$A^T = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Generator Matrix:-

$$G = [I_3 | A]$$

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

The no of ~~words~~ Codes in Z_2^3 is given by $2^3 = 8$ are.

(000) (001) (010) (100), (110), (101), (011), (111)

The Code words.

$E(w) = [w]G$. Where w is the word & G is the generator matrix

$$E(000) = [000]G = [000]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [000000]$$

$$E(100) = [100]\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$= [100111]$$

$E(010) = [010] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 001 \end{bmatrix}$

$= [010010]$

$E(001) = [001] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 001 \end{bmatrix}$

$= [001101]$

$E(110) = [110] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 101 \end{bmatrix}$

$= [110101]$

$E(101) = [101] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 101 \end{bmatrix}$

$= [101010]$

$E(011) = [011] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 001 \end{bmatrix}$

$= [011111]$

$E(111) = [111] \begin{bmatrix} 100 & 111 \\ 010 & 010 \\ 001 & 101 \end{bmatrix}$

$= [111000]$.

Hence the code words are

$C = \{000000, 100111, 010010, 001101, 110101, 101010, 011111, 111000\}$.

(b) since 2nd and 6th column of H are identical which are 010. Hence H does not correct Errors.

(7) An Encoding function $E: Z_2^3 \longrightarrow Z_2^4$ is defined by the generator matrix.

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

(a) Find the set of all code words assigned by E.

(b) Determine the associated Parity check-Matrix generated.

Soln:-

(a) $E(000) = [000]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [0000]$

$E(100) = [100]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [1001]$

$E(010) = [010]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [0101]$

$E(001) = [001]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [\quad]$

$E(110) = [110]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [1100]$

$E(011) = [011]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [0110]$

$E(101) = [101]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [0010]$

$E(111) = [111]\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

$= [1111]$

Hence the code words are

$C = \{0000, 0011, 0101, 1001, 1100, 0010, 0110, 1111\}$

(b) The Parity check matrix associated with that

$H = [A^T | I] = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\therefore H = [1111]$